

Gael Berlanga Boemare

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Education

University of Cambridge

Cambridge, United Kingdom

Integrated Master of Engineering - MEng, Engineering Tripos

Sep. 2022 – Jul. 2026

Master Research Project: Bayesian inference for non-linear state-space models.

Bachelor grade: First Class Honours.

Master grade: Merit.

British School of Brussels

Brussels, Belgium

A-levels: A* A* A* in Mathematics, Physics, Further Mathematics

Sep. 2020 – Jul. 2022

Awards:

- European Statistics Competition 2020. 1st place, team of 3, 17,000 students, 16 countries.
- Belgian Physics Olympiad 2021. 3rd place.

Experience

Quant Trading Intern, Da Vinci Derivatives

Jul. 2025 - Aug. 2025

- Researched relationships between crypto asset baskets.
- Designed a mid-frequency trading strategy on crypto assets using Principal Component Analysis.
- Traded volatility gaps between Nvidia and QQQ options.

Data Science Intern, Galeio

Jul. 2024 - Aug. 2024

- Created vector databases by utilizing Meta's DinoV2 model to generate embeddings from small patches of large satellite images.
- Achieved one-shot object detection by performing searches through vector databases using cosine similarity.
- Developed data annotation software by selecting zero-shot detected objects with Meta's Segment Anything Model, doubling speed.

Data Science Intern, Desbrest Institute of Epidemiology and Public Health

Dec. 2023 - Jan. 2024

- Developed algorithms for automated pulmonary health analysis from exhaled gases using R.
- Extracted meaningful data from curves of exhaled gases by fitting non-linear models.
- Performed principal component analyses followed by clustering algorithms to classify patients.
- Conducted a study linking coastal weather to stroke cases using R.
- Utilised multiple regression techniques, notably Lasso, to extract meaningful variables.

Engineering Intern, AGRO – OLEUM INGENIERIA S.L.

Jul. 2023 - Aug. 2023

- Collaborated with a team of engineers to aid in the production and development of warehouses.
- Designed and edited plans using AutoCAD from data extracted from modelling software.

Competitions and Projects

Algotrade Hackathon — Negative Eevee:

- Won 1st place and 5,000 EUR prize money with a team of 4 in a 24-hour high-frequency trading competition.
- Built a cross-exchange arbitrage algorithm in Rust to exploit pricing inefficiencies.
- Beat the 100 ms data feed by polling order books adaptively depending on volatility and spread whilst respecting rate limits.
- Prototyped a leader-lagger strategy.

IMC Trading Competition Prosperity 2:

- Placed top 1% out of 10,000 teams in a team of 4 in simulated trading competition.
- Developed Strategies using the Black-Scholes equation, z-score of spreads, moving averages, and other indicators.

Poker bot using Counterfactual Regret Minimization:

- Produced a bot that can play Kuhn and Texas Hold 'em poker variants.
- Trained using counterfactual regret minimization with abstraction of the game.

Languages: French, Spanish, and English. All native proficiency.